The equality constraints in matrix form are

$$Ax = b, \text{ where } A \in \mathbb{R}^{2\times 4},\ b \in \mathbb{R}^2,\ x \in \mathbb{R}^4,$$
$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{pmatrix}, b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

We have a problem with $n = 4$ variables and $m = 2$ equality constraints. We consider the basic solution where x_1 and x_2 are in the basis:

$$B = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}; B^{-1} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 0 \\ 1 \end{pmatrix}; x = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

The basic entries of the basic direction corresponding to the non basic variable x_3 is calculated as

$$-B^{-1}A_3 = -\begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{2}{3} \\ -\frac{1}{3} \end{pmatrix}.$$

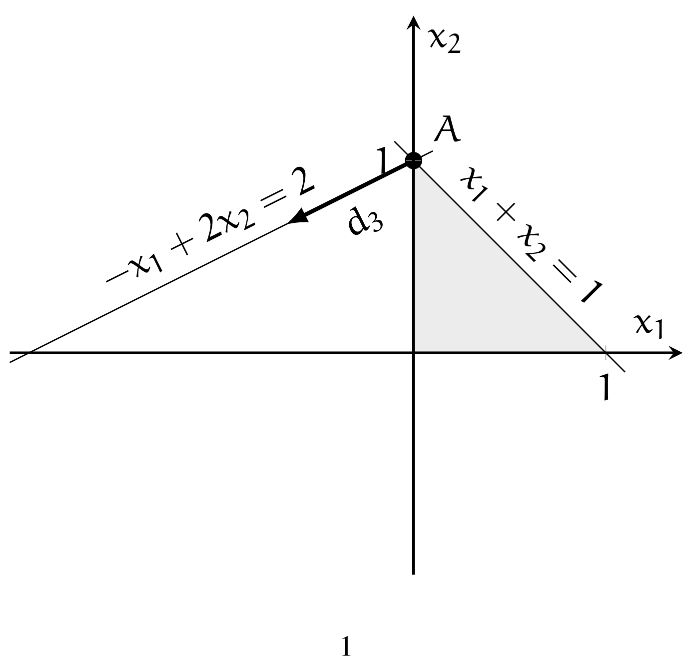
Therefore, the basic direction is

$$d_3 = \begin{pmatrix} -\frac{2}{3} \\ -\frac{1}{3} \\ 1 \\ 0 \end{pmatrix}.$$

Moving along d_3, we obtain

$$x^+ = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} -\frac{2}{3} \\ -\frac{1}{3} \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{2}{3}\alpha \\ 1 - \frac{1}{3}\alpha \\ \alpha \\ 0 \end{pmatrix},$$

which is infeasible for any $\alpha > 0$. x_1 becomes negative as long as we start following the direction. Therefore, the basic direction d_3 is infeasible. It is represented in the following figure.



The basic entries of the basic direction corresponding to the non basic variable x_4 is calculated as

$$-B^{-1}A_4 = -\begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{3} \\ -\frac{1}{3} \end{pmatrix}.$$

Therefore, the basic direction is

$$d_4 = \begin{pmatrix} \frac{1}{3} \\ -\frac{1}{3} \\ 0 \\ 1 \end{pmatrix}.$$

Moving along d_4, we obtain

$$x^+ = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \alpha \begin{pmatrix} \frac{1}{3} \\ -\frac{1}{3} \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{3}\alpha \\ 1 - \frac{1}{3}\alpha \\ 0 \\ \alpha \end{pmatrix},$$

which is feasible for $0 \leq \alpha \leq 3$. Therefore, the basic direction d_4 is feasible. It is represented in the following figure.